

CAMPUS PLACEMENT AND SALARY PREDICTION: LEVERAGING MACHINE LEARNING FOR ENHANCED EMPLOYABILITY

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ABSTRACT

Analyzing extensive campus placement data to extract insights using advanced machine learning techniques, identifying influential factors impacting placement success. This investigation aims to bridge understanding gaps and formulate targeted strategies for optimizing the placement process, presenting a critical problem requiring strategic intervention.

INTRODUCTION

Higher education focuses on predictive models for campus placement and salary projections. Machine learning algorithms, especially ensembles, improve placement rates by balancing datasets. Our research predicts first salaries and job placements accurately, refining placement processes for undergraduates. We're developing a user-friendly program and website for prediction checks, continuously improving algorithms for accuracy.

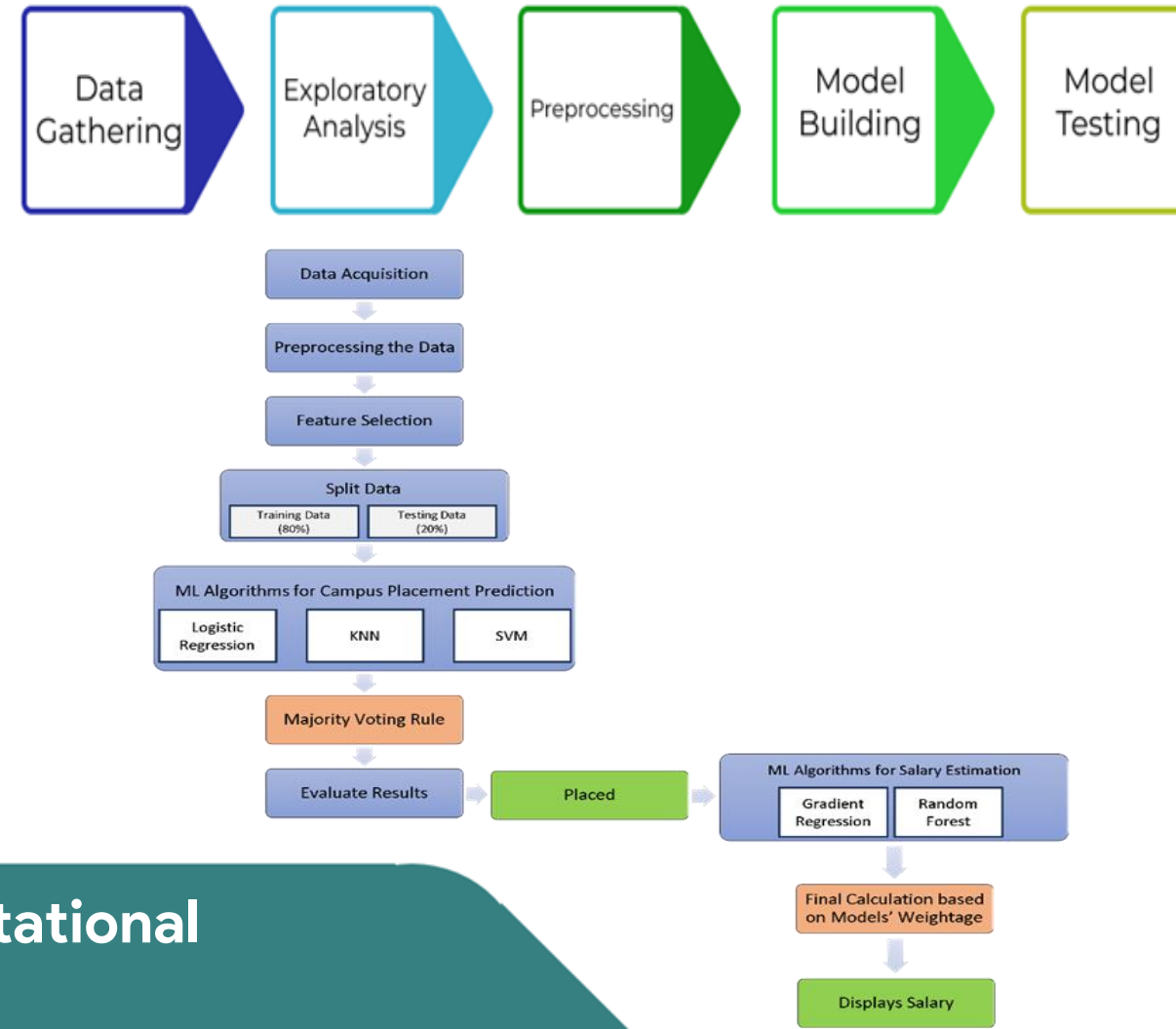
LITERATURE REVIEW

1. **Sachin et al:** Utilized AdaBoost Classifier post data preprocessing, emphasizing strong correlations of "Status" with academic scores and work experience, achieving higher accuracy.
2. **Animesh et al:** Addressed engineering student placement challenges with an integrated model, outperforming Logistic Regression and Support Vector Machine, indicating wider applicability in engineering domains.
3. **Vikas et al:** Employed Decision Trees, SMOTE, Fuzzy Rules, Clustering Techniques, and Naïve Bayes, achieving high accuracy and suggesting future feature expansion.
4. **Nikhil et al:** Employed various techniques to predict computer science student placements, offering insights for students and institutions.
5. **Shahane:** Applied Logistic Regression and other methods, achieving high accuracy and proposing deep learning and cross-validation for further improvement.
6. **Jakub et al:** Utilized Recurrent Neural Networks with NAdam, demonstrating high accuracy and proposing early-age success prediction.

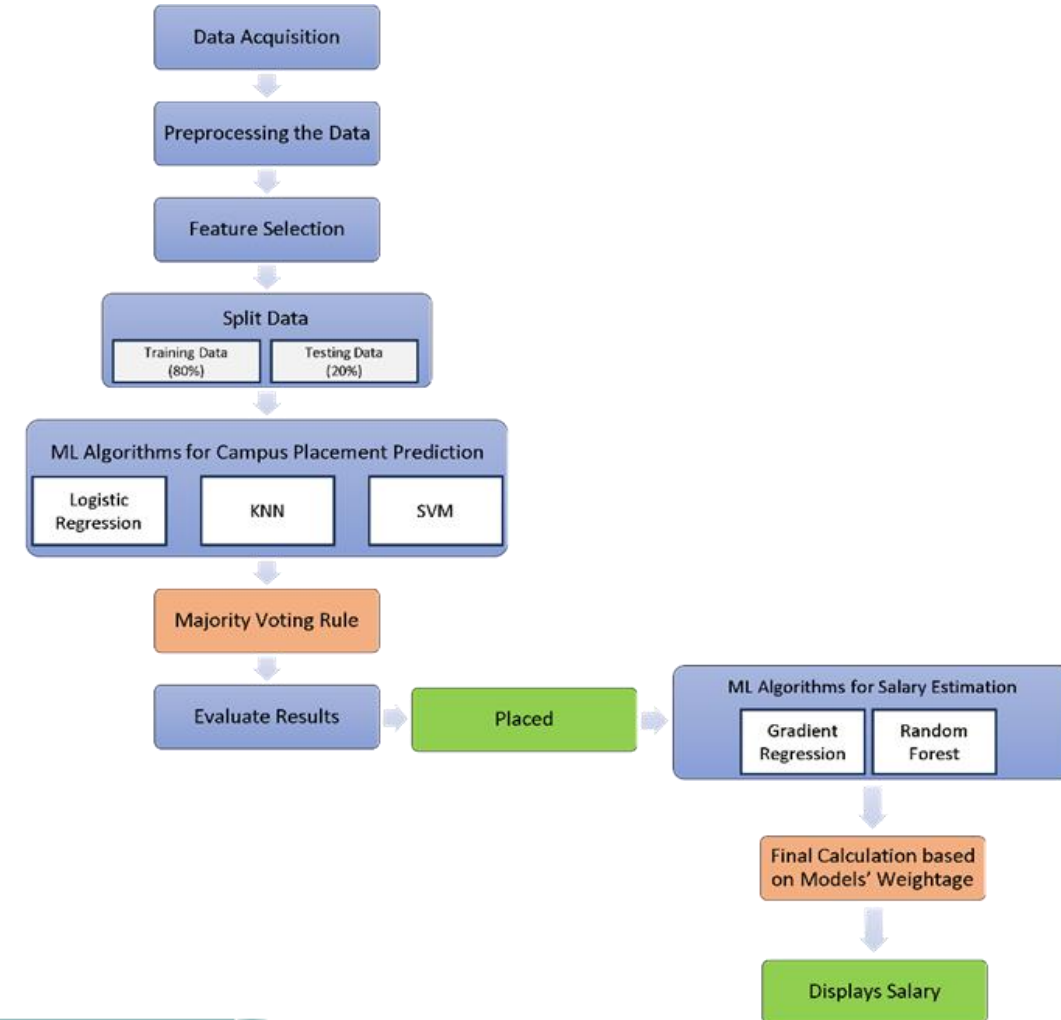
And More...

PROPOSED METHODOLOGY

- Select models including Logistic Regression, KNN, SVM for binary outcome prediction which predicts whether the student gets placed or not.
- A majority voting rule ensures accurate prediction by combining outputs from classification models.

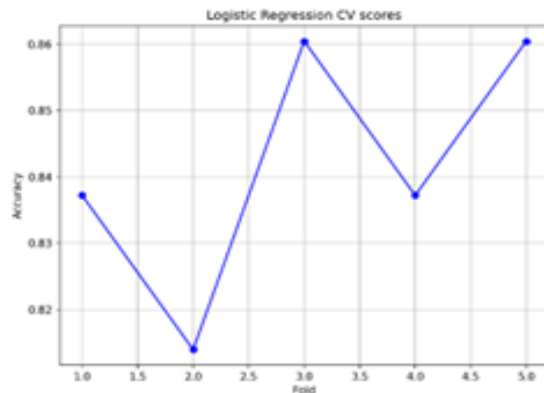


- If the student is estimated to get placed, then salary estimation is done using Random Forest and Gradient Regression.
- Random Forest collaborates with other models, using decision trees for accuracy, while Gradient Regression utilizes XGBoost for salary prediction.
- Weights are assigned based on individual accuracies: Gradient Regression (65.05%), Random Forest (34.95%), based on which the estimated salary is displayed

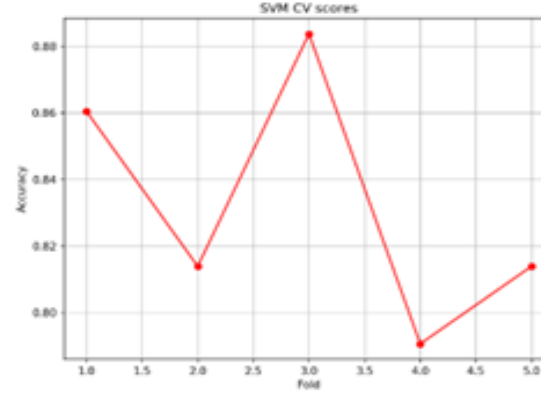


RESULTS AND DISCUSSION

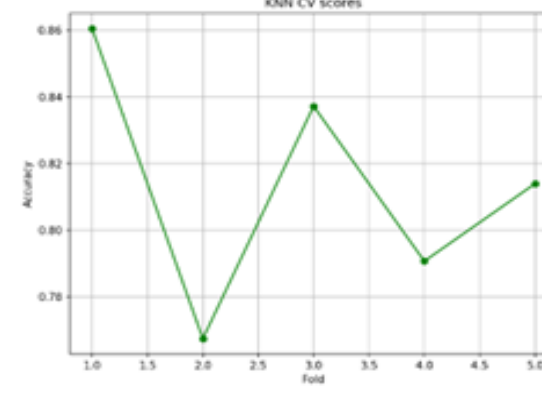
- Various machine learning models, including Logistic Regression, KNN, SVM, Gradient Regression, and Random Forest, were implemented for predicting college placement outcomes and estimating salaries.
- For the classification models, the performance metrics revealed high accuracy for Logistic Regression (84%), followed by SVM (83%) and KNN (81%).



(a) Logistic Regression Model

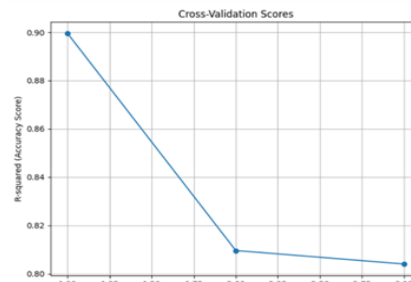


(b) SVM Model



(c) KNN Model

- The models which were leveraged for salary estimation: Gradient Regression achieved 79% accuracy, while Random Forest yielded 68%.

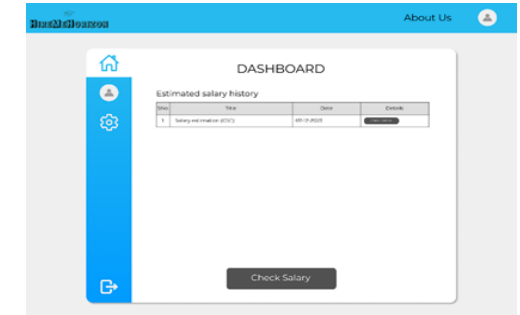


(a) Gradient Regression Model

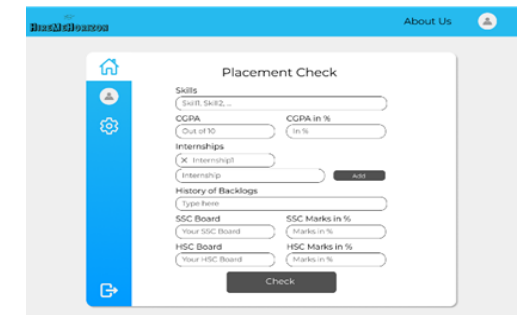


(b) Random Forest Model

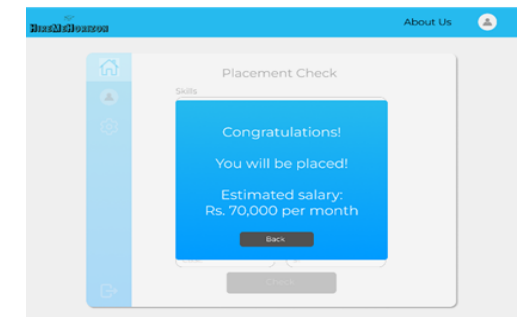
- A custom-built website interface was developed to facilitate user interaction with the prediction models, providing a dashboard for monitoring results and a submission form for placement prediction.



(a) Dashboard



(b) Entering Details for Placement Prediction



(c) Prediction Results with Estimation Results

CONCLUSION & FUTURE SCOPE

This paper introduces a campus placement and salary prediction system utilizing data mining techniques. It explores classifier algorithms' utility for HEIs placement cells, predicting student placements, and refining training processes. Results highlight the model's adaptability and accuracy with other classifiers despite specific datasets.

Future work entails expanding datasets with additional attributes, combining machine learning models across engineering branches, and developing user-friendly Android and desktop/GUI prediction apps. User-entered details will enhance dataset and model training, aiming to improve predictive capabilities in campus placements despite limitations in dataset size and reliance on statistical data.

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THANK YOU